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Coil component

Abstract

A coil component includes a support member, an internal coil supported by the support member and including a plurality of coil patterns, and external electrodes connected to the internal coil and including a first layer in contact with the internal coil and a second layer disposed on the first layer. The second layer is a composite layer including a conductive material and a resin. The support member includes first and second surfaces facing the external electrodes, respectively, and one or more of at least a portion of the first surface and at least a portion of the second surface are configured as cut surfaces.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application is the continuation application of U.S. patent application Ser. No. 16/031,639 filed on Jul. 10, 2018, which claims the benefit of priority to Korean Patent Application No. 10-2017-0124287 filed on Sep. 26, 2017 in the Korean Intellectual Property Office, the disclosure of which is incorporated herein by reference in its entirety.

BACKGROUND

1. Field

(1) The present disclosure relates to a coil component, and more particularly, to a power inductor.

2. Description of Related Art

(2) An inductor, a coil electronic component, is a typical passive element constituting an electronic circuit together with a resistor and a capacitor to cancel noise. Such an inductor, based on electromagnetic properties thereof, is used to configure a resonant circuit that amplifies a signal of

a specific frequency band, a filter circuit, and the like, in combination with a capacitor.

(3) In recent years, metal-based power inductors using amorphous metal or crystalline metal materials have been widely applied to mobile devices due to excellent DC bias characteristics and power conversion efficiency characteristics. In the future, metal-based power inductors are expected to gradually be expansively employed in the industrial and electric fields, and thus, power inductors satisfying high levels of reliability, for example, good contact between internal coils and external electrodes, are required.

SUMMARY

(4) An aspect of the present disclosure may provide a coil component in which contact between an internal coil and external electrodes connected to the internal coil is improved.

(5) According to an aspect of the present disclosure, a coil component may include a support member; an internal coil supported by the support member and including a plurality of coil patterns; and external electrodes connected to the internal coil and including a first layer in contact with the internal coil and a second layer disposed on the first layer. Here, the second layer is a composite layer including a conductive material and a resin. The support member includes first and second surfaces facing the external electrodes, respectively, and one or more of at least a portion of the first surface and at least a portion of the second surface are configured as cut surfaces non-parallel with major surfaces of the support members.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) The above and other aspects, features and other advantages of the present disclosure will be more clearly understood from the following detailed description taken in conjunction with the accompanying drawings, in which:

(2) FIG. 1 is a schematic perspective view of a coil component according to an exemplary embodiment in the present disclosure;

(3) FIG. 2 is a schematic cross-sectional view taken along line I-I' in FIG. 1 according to an embodiment of the present disclosure;

(4) FIG. 3 is a schematic cross-sectional view taken along line I-I' in FIG. 1 according to another embodiment of the present disclosure, and

(5) FIG. 4 is a schematic cross-sectional view of a region "A" of FIG. 2 according to a modification.

DETAILED DESCRIPTION

(6) Exemplary embodiments of the present disclosure will now be described in detail with reference to the accompanying drawings.

(7) Hereinafter, a coil component **100** according to an exemplary embodiment in the present disclosure will be described, but is not limited thereto.

(8) FIG. 1 is a schematic perspective view of a coil component **100** according to an exemplary embodiment in the present disclosure, and FIG. 2 is a schematic cross-sectional view taken along the line I-I' of FIG. 1.

(9) Referring to FIGS. 1 and 2, a coil component **100** according to an exemplary embodiment in the present disclosure includes a body **1** and external electrodes **21** and **22** disposed on outer surfaces of the body **1**.

(10) The body **1** shows an appearance of a coil component **100** and includes upper and lower surfaces opposing each other in the thickness direction T, first and second end surfaces opposing each other in the length direction L, and first and second side surfaces opposing each other in the width direction W, having a substantially hexahedral shape, but is not limited thereto.

(11) The body **1** includes a magnetic material **11**. For example, the body **1** may be formed to be

filled with ferrite or a metal-based soft magnetic material. The ferrite may include a known ferrite such as Mn—Zn ferrite, Ni—Zn ferrite, Ni—Zn—Cu ferrite, Mn—Mg ferrite, Ba ferrite or Li ferrite. The metal-based soft magnetic material may be an alloy including at least one selected from the group consisting of Fe, Si, Cr, Al and Ni, for example, the Fe—Si—B—Cr-based amorphous metal particles but is not limited thereto. The metal-based soft magnetic material may have a particle diameter of 0.1 μm to 20 μm and may be included in a state of being dispersed in a polymer such as an epoxy resin or polyimide.

(12) In the body **1**, a support member **12** sealed by the magnetic material **11** is disposed. The support member **12** serves to facilitate formation of the internal coil **13** on an upper surface or a lower surface thereof and appropriately supports the internal coil **13**. The support member **12** may be formed as a thin plate having insulation properties as a whole. For example, the support member **12** is a central core of a copper clad laminate (CCL) or a printed circuit board (PCB) but is not limited thereto. The support member **12** may have a thickness (i.e., a maximum thickness of the support member) sufficient for supporting the internal coil **13**. For example, the thickness may be about 60 μm . However, when it is considered to extend a utilization field to industrial or electric field product family, it is preferable to employ a support member **12** having a thickness of about 100 μm and it is also possible to employ a support member **12** having a glass transition point (T_g) ranging from 250° C. to 350° C., i.e., having T_g characteristics of a relatively high temperature range.

(13) An upper coil **13a** and a lower coil **13b** are disposed on upper and lower surfaces of the support member **12**, respectively. The upper coil **13a** and the lower coil **13b** form the internal coil **13** as a whole. The upper and lower coils **13a** and **13b** are electrically connected to each other through a via electrode V formed in the support member **12**. The support member **12** may further include a through hole (H) in a central portion thereof in addition to a hole for the via electrode V filled with a conductive material. The through hole is filled with a magnetic material, whereby magnetic permeability of the coil component **100** may be significantly improved. Although not shown, the via electrode V may be provided in plurality, and here, the number of the via electrodes V is not limited. The configuration of a plurality of via electrodes V is to prevent an open defect of a via. Even a single via electrode V may be sufficient for an electrical connection without a problem, but the configuration of a plurality of via electrodes V may effectively prevent an open defect without a substantial change in electrical characteristics.

(14) Referring to FIGS. **1** and **2**, the support member **12** includes a first surface **121** and a second surface **122** facing the external electrodes **21** and **22**, respectively. The first surface **121** and/or the second surface **122** of the support member **12** are cut surfaces. Here, the fact that the first surface **121** and/or the second surface **122** of the support member **12** are cut surfaces indicates that removing at least a portion from the thin plate-like support member **12** during formation of the coil component **100** of the present disclosure is essentially included. The removing of at least a portion of the support member **12** is not limited to a specific method.

(15) For example, at least a portion of the first surface **121** and/or the second surface **122** of the support member **12** may be removed using a CO.sub.2 laser. As a result, the first surface **121** and/or the second surface **122** may have a thickness smaller than a maximum thickness of the support member **12**, and the first surface **121** and/or the second surface **122** may each be reduced in thickness toward the first external electrode **21** and the second external electrode **22**, but is not limited thereto.

(16) A removal shape of the first and second surfaces **121** and **122** of the support member **12** is not particularly limited and may be appropriately selected by a person skilled in the art. For example, as illustrated in FIGS. **1** and **2**, the support member **12** may be removed in a predetermined ratio in the length direction L, but without being limited thereto, the first and second surfaces **121** and **122** may be appropriately varied to have a concave or convex curved shape. Also, the cut surface **121** and **122** may be non-parallel to the major surfaces of the support member **12**.

(17) Since the first surface **121** and/or the second surface **122** of the support member **12** are formed as cut surfaces, the first external electrode **21** or the second external electrode **22** facing the first surface **121** or the second surface **122** and the support member **12** are prevented from being in direct contact with each other. That is, since first layers **211** and **221**, which are the innermost surfaces of the first and second external electrodes **21** and **22**, do not contain a resin and are formed of a single metal or an alloy, the first layers **211** and **221** and the support member **12** formed of a material (e.g. insulation characteristic material) which does not have great bonding strength with the first layers **211** and **221** are prevented from being in direct contact with each other to degrade mutual adhesion. A space between the first layer **211** or **221** and the first surface **121** or the second surface **122** of the support member **12** may be filled with a magnetic material of the body **11**. The magnetic material of the body **11** may certainly improve magnetic permeability and prevent direct contact of the first layers **211** and **221** with the support member **12**.

(18) An insulating layer **31** including a material having insulating properties is disposed on surfaces of the first and second surfaces **121** and **122**. There is no space for arrangement of a separate insulating layer **31** on both end surfaces of the related art support member opposing each other in the length direction L of the conventional support member. In contrast, in the coil component **100** according to an exemplary embodiment in the present disclosure, since the first and second surfaces **121** and **122** of the support member **12**, as well as the through hole at the central portion of the support member **12**, are formed after the coils **13** are formed on the upper and lower surfaces of the support member **12** (e.g., through plating), the surfaces of the first and second surfaces **121** and **122** of the support member **12** may be coated with the insulating layer **31** during a process of insulating the coil **13** subsequently applied after the plating process. A specific thickness is not limited and the insulating layer **31** may have the substantially same thickness as an insulation thickness formed on the coil patterns **13**. Also, a material of the insulating layer **31** is not limited. For example, the insulating layer **31** may be formed of a perylene resin capable of forming a uniform insulating layer **31** through chemical vapor deposition (CVD) but is not limited thereto. Also, as illustrated in FIG. 2, since the insulating layer **31** is formed through the same process as the insulating layer **31** for insulating coil patterns **13** therebetween, the insulating layer **31** naturally continuously extends, as the insulating layer **31** for insulating the coil patterns **13**, to the insulating layer **31** disposed on the coil patterns **13**.

(19) Referring to the first and second external electrodes **21** and **22**, respectively facing the first and second surfaces **121** and **122** of the support member **12**, the first and second external electrodes **21** and **22** include the first layers **211** and **221** and second layers **212** and **222** disposed thereon, respectively. Since descriptions of the first external electrode **21** may be applied to the second external electrode **22** as is, redundant descriptions of the second external electrode **22** will be omitted for the purposes of description. The first layer **211** and the second layer **212** of the first external electrode **21** are formed of materials having different characteristics. A greatest difference between the first layer **211** and the second layer **212** is that the first layer **211** does not contain a resin while the second layer **212** includes a resin with a conductive material dispersed therein. For example, the first layer **211** may include Cu and/or Ni, while the second layer **212** may be formed of a silver (Ag)-epoxy composite but is not limited thereto. There is no limitation in a method of forming the first layer **211** on the first and second end surfaces of the body **11**, respectively. A method may be appropriately selected by a person skilled in the art in consideration of process requirements and required characteristic values. For example, a plating process, a process of applying a metal paste, or a process of depositing by sputtering may be utilized. Since the second layer **212** is formed of a copper-epoxy composite, both improvement of conductivity of the external electrode **21** and improvement of molding characteristics may be realized, while a bonding force with respect to a magnetic material and a conductive material constituting the internal coil **13** may be relatively lowered. Here, since the first layer **211** is formed of only a single metal or alloy without a resin is interposed as a buffer layer between the second layer **212** and the body **11**, contact

reliability between the body **11** and the external electrodes **21** and **22** may be improved and contact resistance may be lowered.

(20) The second layer **212** may be formed to cover the entire surface of the first layer **211**, and a third layer **213** may be additionally disposed on a surface of the second layer **212** to include at least one of Ni and Sn. The third layer **213** may be configured as a layer for facilitating soldering, or the like, when the coil component **100** is mounted.

(21) A half of a maximum thickness T_s of the support member **12** may be controlled to be equal to or greater than maximum thicknesses T_{e1} and T_{e2} of the first layers **211** and **221** of the first and second external electrodes **21** and **22**, respectively. If the maximum thickness of the first layer **211** has a value larger than the half of the maximum thickness of the support member **12**, it is not possible to reduce the thickness of the external electrodes **21** and **22** including the second layer **212** as well as the first layer **211**, and as a result, miniaturization of the coil component **100** may not be achieved.

(22) FIG. 3 shows a modification of the inset shown in FIG. 2 according to another embodiment of the present disclosure. In this embodiment, the first and second surfaces **121** and **122** are formed to be slant surfaces, spacing the end of the support member **12''** apart from the external electrodes **21** and **22**. Also, the end of the support member **12''** having the first or second surface **121** or **122** may contact a central region of the coil pattern **13** contacting the external electrodes **21** and **22**.

(23) FIG. 4 is a schematic cross-sectional view according to a modification of the region A of FIG. 2. Compared with FIG. 2, FIG. 4 includes the substantially same components, except that shapes of the first and second surfaces **121** and **122** of a support member **12'** in the region A are different, and thus, for the purposes of description, descriptions other than the difference in components between the coil component **100** of FIG. 2 and the coil component **100** of FIG. 4 will be omitted and the same reference numerals will be used for the same components. Also, descriptions of the first surface **121** of the support member **12'** may also be applied as is to the second surface **122**, and thus, only the first surface **121** of the support member **12'** will be described and redundant descriptions of the second surface will be omitted.

(24) Referring to FIG. 4, the support member **12'** has a substantially uniform thickness in relation to the length direction L of the body **11**, which is different from the support member **12** of the coil component **100** of FIG. 2 in which the substantially uniform thickness is maintained in relation to the length direction L of the body **11** and reduced in the first and second surfaces **121** and **122** in relation to the length direction L of the body **11**. A thickness formed by the first surface **121'** of the support member **12'** is substantially equal to a thickness of the support member **12'** formed in the other position of the support member **12'**, while a total length of the support member **12'** extending in the length direction L of the body **11** is relatively short. This means that the first surface **121'** is formed to be substantially perpendicular to a plurality of coil patterns **13** of the internal coil **13** such as a lead pattern of the internal coil **13**, and is spaced apart from the first layer **211** of the external electrodes **21** and **22** at a predetermined distance. Here, a person skilled in the art may appropriately selectively remove a portion of the support member **12'** by the predetermined distance, and in this case, the portion of the support member **12'** may be removed to the extent that the upper coil **13a** disposed on an upper surface of the support member **12'** and the lower coil **13b** disposed on a lower surface of the support member **12'** are sufficiently directly connected by a via electrode inside the support member **12'**, while the remaining portions of the support member **12'** appropriately support the internal coil **13** supported thereby.

(25) Except for the above descriptions, redundant descriptions of the coil component **100** according to the exemplary embodiment in the present disclosure described above will be omitted.

(26) According to the coil component **100** described above, in the coil component **100** having the external electrodes **21** and **22** including the metal-resin composite layer as at least one layer, the overall thickness of the external electrodes **21** and **22** is reduced, while solving the problem of adhesion reliability between the body and the external electrodes, to follow the trend of

miniaturized electronic components.

(27) As set forth above, according to exemplary embodiments of the present disclosure, the coil component **100** having improved reliability and a low Rdc value by improving contact characteristics between the internal coil **13** and the external electrodes **21** and **22** may be provided.

(28) While exemplary embodiments have been shown and described above, it will be apparent to those skilled in the art that modifications and variations could be made without departing from the scope of the present disclosure as defined by the appended claims.

Claims

1. A coil component comprising: a body comprising a magnetic material; a support member disposed in the body; an internal coil disposed on the support member and including a body pattern including a plurality of turns and a lead pattern connected to an outermost turn of the body pattern; an insulating layer disposed on the internal coil; and external electrodes disposed on the body and including a first layer contacting the lead pattern and a second layer disposed on the first layer, wherein the support member includes a support portion contacting a lower surface of the body pattern, and an extended portion contacting one region of a lower surface of the lead pattern and spaced apart from the first layer of each of the external electrodes in a direction parallel to the support member by the magnetic material of the body, and the insulating layer extends continuously on a surface of the internal coil and a surface of the support member that is not in contact with the internal coil to separate the support member and the internal coil from the magnetic material, and the magnetic material encapsulates the support member, the internal coil and the insulating layer to form the body.
2. The coil component of claim 1, wherein the first layer is formed of a single metal or an alloy.
3. The coil component of claim 2, wherein the first layer comprises copper (Cu).
4. The coil component of claim 1, wherein the second layer comprises a conductive material and a resin.
5. The coil component of claim 1, wherein a space from the extended portion of the support member to the first layer is filled with the magnetic material.
6. The coil component of claims 1, wherein a thickness of at least a portion of the extended portion is thinner than a thickness of the support portion.
7. The coil component of claim 6, wherein a thickness of the extension portion is reduced in a direction, toward the first layer.
8. The coil component of claim 1, wherein the insulating layer is continuously disposed on the plurality of turns of the body pattern and the lead pattern.
9. The coil component of claim 1, wherein a maximum thickness of the first layer has a value equal to half or less of a maximum thickness of the support member.
10. The coil component of claim 1, wherein the second layer is spaced apart from the magnetic material in the body or the internal coil by a predetermined distance.
11. The coil component of claim 1, wherein the first layer does not comprise a resin.
12. The coil component of claim 1, wherein the internal coil comprises an upper coil disposed on an upper surface of the support member, and a lower coil disposed on a lower surface of the support member, wherein the upper and lower coils are electrically connected to each other by one or more vias passing through the support member.
13. The coil component of claim 1, further comprising: a third layer disposed on a surface of the second layer, wherein the third layer includes at least one of Ni and Sn.
14. A coil component comprising: a body comprising a magnetic material; a support member disposed in the body; an internal coil disposed on the support member and including a body pattern including a plurality of turns and a lead pattern connected to an outermost turn of the body pattern; an insulating layer contacting disposed over the internal coil and contiguously extending to at least

a portion of a side surface of the support member; and external electrodes disposed on the body and including a first layer contacting the lead pattern and a second layer disposed on the first layer, wherein the support member includes a support portion contacting a lower surface of the body pattern, and an extended portion contacting one region of a lower surface of the lead pattern and spaced apart from the first layer of each of the external electrodes in a direction parallel to the support member by the magnetic material of the body, the insulating layer extends continuously on a surface of the internal coil and a surface of the support member that is not in contact with the internal coil to separate the support member and the internal coil from the magnetic material, the magnetic material encapsulates the support member, the internal coil and the insulating layer to form the body, and a thickness of at least a portion of the extended portion is thinner than a thickness of the support portion.

15. The coil component of claim 14, wherein a thickness of the extension portion is reduced in a direction, toward the first layer.

16. The coil component of claim 14, wherein a space from the extended portion of the support member to the first layer is filled with the magnetic material.

17. The coil component of claim 14, wherein the second layer comprises a conductive material and a resin.
